

## Extra credit problems

Math 471

0. Find a mistake or misprint in the book. (The score depends on the type of mistake.)
1. Show that a composition of three reflections in the sides of a nondegenerate triangle does not have a fixed point.
2. Given distinct points  $A$ ,  $B$  and  $X$  make a ruler-and-compass construction of another point  $Y \in (AX)$  such that  $AY + BY = AX + BX$  and explain why it works. (You may play with the GeoGebra applet [apfzzybw](#).)
3. Let  $m$  be an indirect motion of the plane. Show that the midpoints of all line segments  $[X m(X)]$  lie on one line.